UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 006 · Particular integral Type 4 ($e^{ax}V$) and Type 5 (xV) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S006·5	Unit 2 deck 006 ("Particular integral of Type 4"), slide 5 (rendered .pptx)
T006 00:04:12	Unit 2 AV-summary transcript "Particular integral of standard functions – Type 4" (17 min), segment start
T006b 00:12:47	Unit 2 AV-summary transcript "Particular integral – Type 3 and Type 4" (bucket 006, 15.5 min); only its Type 4 part (00:09:55–00:15:25) is cited here. The class 005 pack cites the same file's Type 3 part as T006a.
CW U2 P18	Unit 2 class-work problem 18
ESA Jan-24 s3 2c	Past paper: UE23MA141A Jan 2024 set 3, question 2.c

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1 The Type 4 rule: $X = e^{ax}V$ (exponential shift)

Definitions

Type 4 right-hand side. $X = e^{ax}V$, where V is a function of x (in the problems: $\sin bx$, $\cos bx$, a polynomial, or a product such as $x\cos x$). S006·3 · S006·4 · S006·10 · T006 00:00:00

Exponential shift. Moving e^{ax} from the right of $\frac{1}{f(D)}$ to its left while replacing D by $D + a$: "e^{ax} can be taken outside the operator by replacing D by $D + a$ in $f(D)$". S006·4 · T006 00:01:00

a. The coefficient of x in the exponential factor: $a = 1$ for e^x, $a = 3$ for e^{3x}, $a = -1$ for e^{-x}. T006 00:01:18 · T006b 00:12:13

$f(D + a)$. The shifted operator: $f(D)$ with every D replaced by $D + a$, then expanded and simplified (e.g. $D^2 - 2D + 4 \to (D+1)^2 - 2(D+1) + 4 = D^2 + 3$). S006·6 · T006 00:03:41

Statements in the class material

> *"Type (iv); When $X = e^{ax}V$, where V is a function of x:* $\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$*."* S006·4

> *"When $X = e^{ax}V$, V being a function of x, the particular integral is given by* $y_p = \text{P.I.} = \frac{1}{f(D)}e^{ax}V(x) = e^{ax}\frac{1}{f(D+a)}V(x) = e^{ax}[f(D+a)]^{-1}V(x)$ *where $V(x)$ is of some particular form."* S006·10

Key points

1. Recognise Type 4: X is a product with an exponential factor, "e^{ax} into V" ($e^x\cos x$, x^2e^{3x}, $e^{2x}\cos x$). "If x^2 is not there, case 1; if e^{3x} is not there, case 3; both are there, so case 4". ESA ★★ T006 00:00:18 · T006b 00:11:45 · S006·10
2. Write e^{ax} to the **left** of the operator and replace **every** D in $f(D)$ by $D + a$; leave V as it is. ESA ★★ S006·4 · T006 00:01:00 · T006b 00:10:19
3. Expand and simplify $f(D + a)$ before doing anything else: $(D + 1)^2 - 2(D + 1) + 4 = D^2 + 3$; $(D + 1)^2 - 2(D + 1) + 1 = D^2$. ESA ★★ S006·6 · S006·8 · T006 00:03:58 · T006 00:06:23
4. After the shift, the rule is chosen by V alone: $\sin/\cos$ → Type 2; polynomial → Type 3; $f(D + a) = D^k$ → integrate k times. ESA ★★ T006b 00:10:40 · T006b 00:10:53 · T006 00:04:20 · T006 00:12:09
5. In Type 2 after the shift, $D^2 \to -b^2$ uses b from $V = \cos bx$, not a from e^{ax}: $\frac{e^x\cos x}{D^2+3} \to \frac{e^x\cos x}{-1+3}$. ESA ★★ S006·6 · T006 00:04:20 · CW U2 P17
6. The shift handles failure by itself: if a is a root of the AE, $f(D + a)$ has a factor D, and $\frac{1}{D}$ integrates (slide 006·14: $(D - 1)^2 \to D^2$). ESA ★★ S006·14 · T006b 00:14:22
7. "Find the general solution" needs $y = y_c + y_p$ written out: "this is must". ESA ★★ T006 00:09:28 · S006·6 · S006·9

Why?

Why does D become $D + a$?
By the product rule $D(e^{ax}V) = e^{ax}(DV + aV) = e^{ax}(D + a)V$; applying this repeatedly, $f(D)(e^{ax}V) = e^{ax}f(D + a)V$. Reading this backwards for the inverse operator gives E1.1: to undo $f(D)$ on $e^{ax}V$, undo $f(D + a)$ on V.

ADDED FOR CLARITY S006·4

Why does the shift resolve a failure case without a special rule?
If a is a k-fold root of the AE, $f(D) = (D - a)^k\psi(D)$, so $f(D + a) = D^k\psi(D + a)$: the shifted operator has D^k as a factor and the k integrations produce the extra x^k. Slide 006·14 is an example: $(D - 1)^2 \to D^2$.

ADDED FOR CLARITY S006·14 · T006b 00:14:22

Why can Type 2 not be used directly on $e^x \cos x$?
Type 2 replaces D^2 by $-b^2$ because $D^2 \cos bx = -b^2 \cos bx$; but $D^2(e^x \cos x) = -2e^x \sin x$ is not a multiple of $e^x \cos x$. After the shift the operator acts on $\cos x$ alone, where the substitution is valid.
ADDED FOR CLARITY S006·6 · T006 00:03:23

Exam traps

Exam trap. Slide 006·5 writes the CF as $e^x(c_1 \cos\sqrt{3}x + c_1 \cos\sqrt{3}x)$; it is $e^x(c_1 \cos\sqrt{3}x + c_2 \sin\sqrt{3}x)$. ADDED FOR CLARITY S006·5 · T006 00:03:06

Exam trap. Do not leave e^{ax} inside: $\frac{1}{f(D+a)}(e^{ax}V)$ is wrong; the exponential moves to the left and disappears from what the operator acts on. S006·4 · T006 00:01:00

Exam trap. Replace every D, including the D inside $-2D$ and $4D$: $D^2 + 4D + 3 \to (D+2)^2 + 4(D+2) + 3$, not $(D+2)^2 + 4D + 3$. ADDED FOR CLARITY CW U2 P17 · S006·6

Exam trap. Do not put $D = a$ (Type 1) when V is not a constant: that rule is only for $V = 1$.
ADDED FOR CLARITY S006·4

2 Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides

Definitions

$\frac{1}{D^k}$. k successive integrations with respect to x, "without writing the constant of integration". S006·8 · S006·9 · T006 00:06:58 · T006 00:07:16

Key points

1. For $x^m e^{ax}$, take the constant $f(a)$ out of $f(D+a)$ and expand $[1 \pm \varphi(D)]^{-1}$ up to D^m (class 005 rule). ESA ★★ S006·11 · S006·12 · T006b 00:12:47 · T006 00:12:25
2. Stop the series at the degree of the polynomial: for x^2 keep D^2 ("a^2 is not needed, b^2 is needed, $2ab$ is not needed"); for x keep only the D term. ESA ★★ S006·12 · T006b 00:13:25 · T006 00:13:20
3. $\frac{1}{D^2}$ means integrate twice, with no constants (the constants are already in the CF). ESA ★★ S006·8 · T006 00:06:58 · T006 00:07:16
4. With a mixed X (ESA Jan-24 s3 2c), split X, find PI_1 by the shift and PI_2 by Type 2, and add. ESA ★★ ESA Jan-24 s3 2c · CW U2 P18 · T006 00:10:40
5. Leading-term check: the PI of $x^m e^{ax}$ starts with $\frac{x^m e^{ax}}{f(a)}$ when $f(a) \neq 0$ ($\frac{x^2 e^{3x}}{11}$, $-\frac{x^2 e^{-x}}{4}$). ESA ★ ADDED FOR CLARITY S006·12 · CW U2 P18

Exam traps

Exam trap. Slide 006·9 (and T006 00:07:16) integrates $x \sin x$ instead of $x \cos x$; its answer $-e^x(x \sin x + 2\cos x)$ is wrong. $\frac{1}{D^2}(x \cos x) = 2\sin x - x\cos x$, so the PI is $e^x(2\sin x - x\cos x)$. ADDED FOR CLARITY S006·9 · T006 00:07:16

Exam trap. No constants of integration when evaluating $\frac{1}{D}$: "that's very important". T006 00:07:16

Exam trap. Slide 004·17's boxed PI for the $\sin^2 x$ part of ESA Jan-24 s3 2c drops a factor 2; the correct part is $\frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$ (class 004 pack, CW P18). ADDED FOR CLARITY ESA Jan-24 s3 2c · CW U2 P18

3 The Type 5 rule: $X = xV$

Definitions

Type 5 right-hand side. $X = xV$, where V is a function of x (in the problems: $\sin ax$ or $\cos ax$). S006·4 · T006 00:01:28

$f'(D)$. The derivative of the polynomial $f(D)$ with respect to D: for $f(D) = D^2 - 5D + 6$, $f'(D) = 2D - 5$. S006·4 · T006 00:01:28

Statements in the class material

> *"Type (v); When $X = xV$ where V is a function of x:* $\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$." S006·4

Key points

1. Type 5: $\frac{1}{f(D)}(xV) = x\frac{1}{f(D)}V - \frac{f'(D)}{(f(D))^2}V$. Two Type 2 computations: first $\frac{V}{f(D)}$, then $\frac{f'(D)V}{(f(D))^2}$. ESA ★★ S006·4 · T006 00:01:28 · CW U2 P19 · CW U2 P20
2. In $\frac{f'(D)}{(f(D))^2}V$, replace D^2 by $-a^2$ in $(f(D))^2$ after squaring, then rationalise any leftover D. ESA ★★ ADDED FOR CLARITY CW U2 P20
3. Check with the complex shortcut: $x\sin ax = \mathrm{Im}(xe^{iax})$, Type 4 with $a \to ia$ and $V = x$. ESA ★ ADDED FOR CLARITY CW U2 P19

Why?

Why is there a correction term $-\frac{f'(D)}{(f(D))^2}V$?
Multiplying by x does not commute with D: $D(xU) = xDU + U$. So $f(D)$ applied to $x\frac{V}{f(D)}$ gives xV plus the extra $f'(D)\frac{V}{f(D)}$; the correction term removes that extra (E3.2).
ADDED FOR CLARITY S006·4

Exam traps

Exam trap. $\frac{x}{f(D)}V$ is $x \cdot \left(\frac{1}{f(D)}V\right)$, not $\frac{1}{f(D)}(xV)$ (that is what you are trying to find).
ADDED FOR CLARITY S006·4

Exam trap. Square $f(D)$ before substituting $D^2 = -a^2$ is fine, but keep the cross term: $(2 - 5D)^2 = 4 - 20D + 25D^2$, not $4 + 25D^2$. ADDED FOR CLARITY CW U2 P20

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences; 7 distinct / 9 occurrences are PI by operator rules). Type 4 is asked once: Jan-24 set 3 2c, $(D^3 - 2D^2 + D)y = x^2e^{-x} + \sin^2 x$ (6 marks), which is also CW P18 (solved as W2.3). Type 5 (xV) is never asked on its own; it is computed only in CW P19 and P20. Stars: the Type 4 rule and $x^m e^{ax}$ ★★ (one ESA question plus CW P17, P18); Type 5 ★★ (CW only); the complex-exponential shortcut ★ (added for clarity).

2 · Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q2.c (= CW P18) · solved in W2.3

Solve $(D^3 - 2D^2 + D)y = x^2e^{-x} + \sin^2 x$.